

Packet Tracer: propaga una ruta predeterminada en OSPFv2

Tabla de asignación de direcciones

Dispositivo	Interfaz	Dirección IPv4	Máscara de subred	Gateway predeterminado
R1	G0/0	172.16.1.1	255.255.255.0	N/D
	S0/0/0	172.16.3.1	255.255.255.252	
	S0/0/1	192.168.10.5	255.255.255.252	
R2	G0/0	172.16.2.1	255.255.255.0	N/D
	S0/0/0	172.16.3.2	255.255.255.252	
	S0/0/1	192.168.10.9	255.255.255.252	
	S0/1/0	209.165.200.225	255.255.255.224	
R3	G0/0	192.168.1.1	255.255.255.0	N/D
	S0/0/0	192.168.10.6	255.255.255.252	N/D
	S0/0/1	192.168.10.10	255.255.255.252	N/D
PC1	NIC	172.16.1.2	255.255.255.0	172.16.1.1
PC2	NIC	172.16.2.2	255.255.255.0	172.16.2.1
PC3	NIC	192.168.1.2	255.255.255.0	192.168.1.1
Servidor web	NIC	64.100.1.2	255.255.255.0	64.100.1.1

Objetivos

Parte 1. Propagar una ruta predeterminada

Parte 2. Verificar la conectividad

Aspectos básicos

En esta actividad, configurará una ruta predeterminada IPv4 a Internet y realizará la propagación de esa ruta predeterminada a otros routers OSPF. A continuación, verificará que la ruta predeterminada figure en las tablas de enrutamiento corriente abajo y que los hosts ahora puedan acceder al servidor web en Internet.

Instrucciones

Parte 1: Propagar una ruta predeterminada

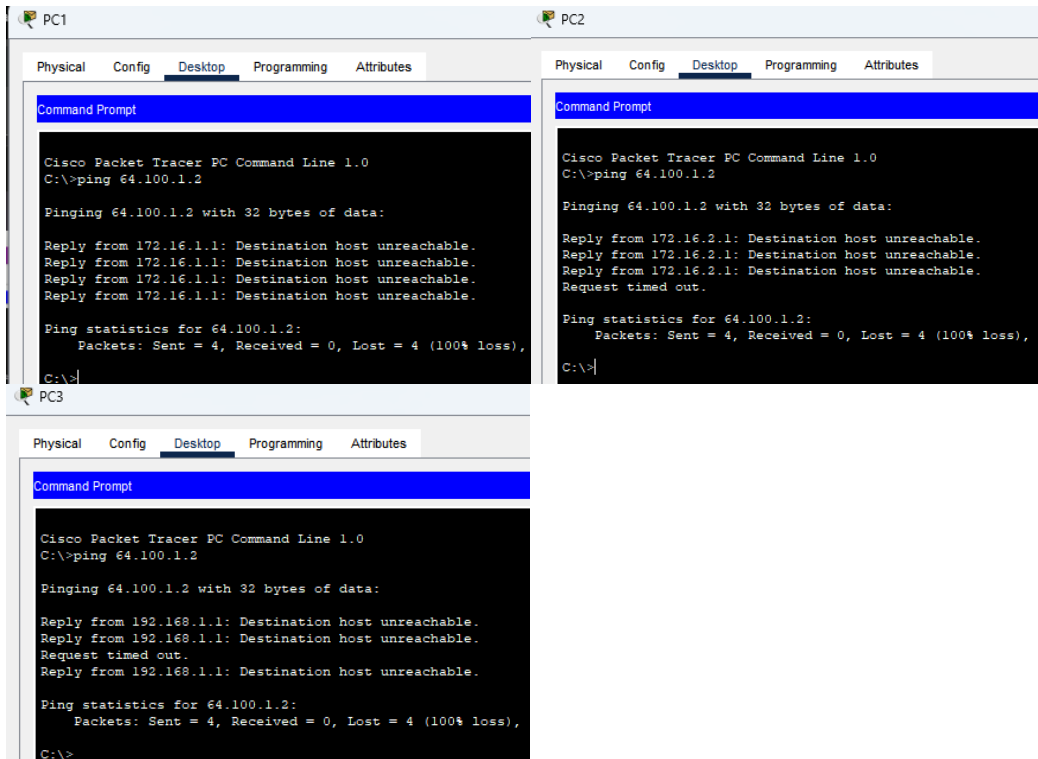
Paso 1: Probar la conectividad al servidor web

- Desde PC1, PC2 y PC3, intente hacer ping a la dirección IP del servidor Web 64.100.1.2.

¿Alguno de los pings tuvo éxito? NO

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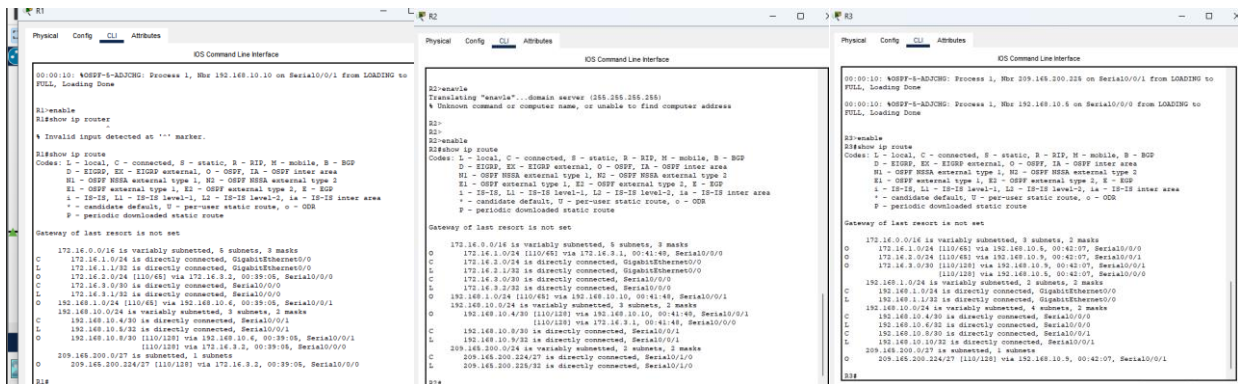
Prueba



¿Qué mensaje recibí y qué dispositivo emitió el mensaje?

host de destino inalcanzable, lo emite R2 por que es que tiene salida a internet.

- b. Examine las tablas de enrutamiento en los routers R1, R2 y R3.



¿Qué instrucción está presente en las tablas de enrutamiento que indica que los pings al servidor Web fallarán?

La puerta de enlace de último recurso no está configurada

Paso 2: Configurar una ruta predeterminada en el R2.

Configure el R2 con una ruta predeterminada conectada directamente a Internet.

R2(config)# **ip route 0.0.0.0 0.0.0.0 Serial10/1/0**

```
R2#conf ter
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#ip route 0.0.0.0 0.0.0.0 s0/1/0
%Default route without gateway, if not a point-to-point interface, may impact performance
R2(config)#
```

Nota: El router dará una advertencia de que si esta interfaz no es una conexión punto a punto, puede afectar el rendimiento. Puede ignorar esta advertencia porque es una conexión punto a punto.

Paso 3: Propagar la ruta en OSPF.

Configure OSPF para propagar la ruta predeterminada en las actualizaciones de enrutamiento OSPF.

```
R2(config)# router ospf 1
R2(config-router)# default-information originate
```

```
R2(config)#router ospf 1
R2(config-router)#default-information originate
^
% Invalid input detected at '^' marker.
R2(config-router)#default-information originate
R2(config-router)#
```

Paso 4: Examinar las tablas de enrutamiento del R1 y el R3.

Examine las tablas de enrutamiento en el R1 y el R3 para verificar que la ruta se haya propagado.

```
R1> show ip route
<output omitted>
Gateway of last resort is 172.16.3.2 to network 0.0.0.0
<output omitted>
O*E2 0.0.0.0/0 [110/1] via 172.16.3.2, 0:00:08, Serial0/0/0
!-----
R3> show ip route
<output omitted>
Gateway of last resort is 192.168.10.9 to network 0.0.0.0
<output omitted>
O*E2 0.0.0.0/0 [110/1] via 192.168.10.9, 0:08:15, Serial0/0/1
```

Prueba

```
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 172.16.3.2 to network 0.0.0.0

R1#
172.16.0.0/16 is variably subnetted, 5 subnets, 3 masks
C 172.16.1.0/24 is directly connected, GigabitEthernet0/0
L 172.16.1.1/32 is directly connected, GigabitEthernet0/0
O 172.16.2.0/24 [110/65] via 172.16.3.2, 00:53:02, Serial0/0/0
C 172.16.3.0/30 is directly connected, Serial0/0/0
L 172.16.3.1/32 is directly connected, Serial0/0/0
O 192.168.1.0/24 [110/65] via 192.168.10.6, 00:53:02, Serial0/0/1
L 192.168.1.0/24 is variably subnetted, 3 subnets, 2 masks
C 192.168.10.4/30 is directly connected, Serial0/0/1
L 192.168.10.5/32 is directly connected, Serial0/0/1
O 192.168.10.8/30 [110/128] via 172.16.3.2, 00:53:02, Serial0/0/0
O 209.165.200.0/27 is subnetted, 1 subnets
O 209.165.200.224/27 [110/128] via 172.16.3.2, 00:53:02, Serial0/0/0
O*E2 0.0.0.0/0 [110/1] via 172.16.3.2, 00:02:13, Serial0/0/0
R1#
```

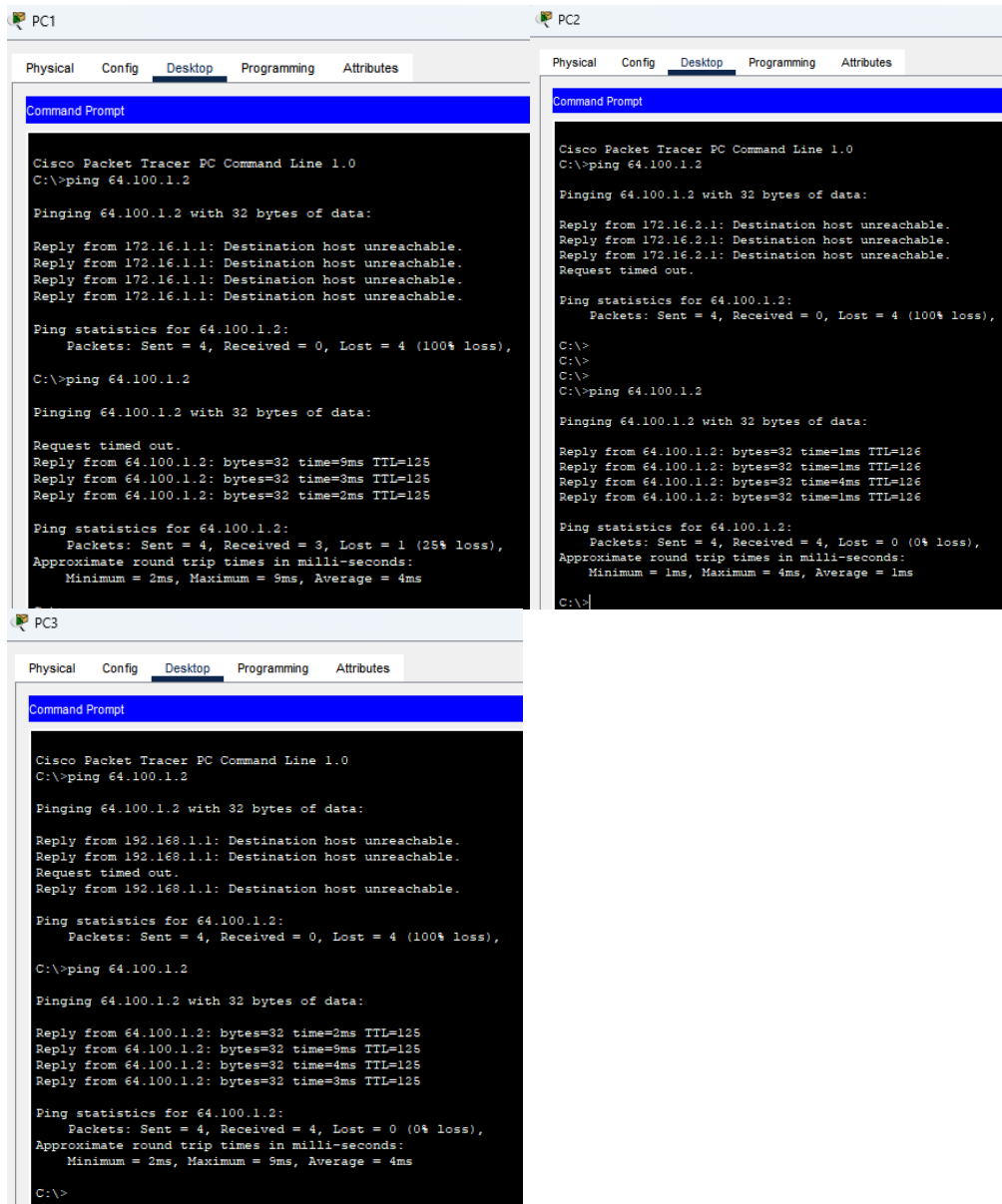
```
R3#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 192.168.10.9 to network 0.0.0.0

R3#
172.16.0.0/16 is variably subnetted, 3 subnets, 2 masks
O 172.16.1.0/24 [110/65] via 192.168.10.6, 00:53:31, Serial0/0/0
O 172.16.2.0/24 [110/65] via 192.168.10.9, 00:53:31, Serial0/0/1
O 172.16.3.0/30 [110/128] via 192.168.10.9, 00:53:31, Serial0/0/1
O 192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.1.0/24 is directly connected, GigabitEthernet0/0
L 192.168.1.1/32 is directly connected, GigabitEthernet0/0
C 192.168.10.4/30 is directly connected, Serial0/0/0
L 192.168.10.6/32 is directly connected, Serial0/0/0
C 192.168.10.8/30 is directly connected, Serial0/0/1
L 192.168.10.10/32 is directly connected, Serial0/0/1
O 209.165.200.0/27 is subnetted, 1 subnets
O 209.165.200.224/27 [110/128] via 192.168.10.9, 00:53:31, Serial0/0/1
O*E2 0.0.0.0/0 [110/1] via 192.168.10.9, 00:02:47, Serial0/0/1
R3#
```

Parte 2: Verificar la conectividad

Verify that **PC1**, **PC2**, and **PC3** can ping the web server.



The image displays three screenshots of the Cisco Packet Tracer PC Command Line interface for PC1, PC2, and PC3. Each window shows the results of a ping command to the IP address 64.100.1.2.

PC1 Command Prompt:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 64.100.1.2

Pinging 64.100.1.2 with 32 bytes of data:

Reply from 172.16.1.1: Destination host unreachable.
Reply from 172.16.1.1: Destination host unreachable.
Reply from 172.16.1.1: Destination host unreachable.
Reply from 172.16.1.1: Destination host unreachable.

Ping statistics for 64.100.1.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 64.100.1.2

Pinging 64.100.1.2 with 32 bytes of data:

Request timed out.
Reply from 64.100.1.2: bytes=32 time=9ms TTL=125
Reply from 64.100.1.2: bytes=32 time=3ms TTL=125
Reply from 64.100.1.2: bytes=32 time=2ms TTL=125

Ping statistics for 64.100.1.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 9ms, Average = 4ms
```

PC2 Command Prompt:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 64.100.1.2

Pinging 64.100.1.2 with 32 bytes of data:

Reply from 172.16.2.1: Destination host unreachable.
Reply from 172.16.2.1: Destination host unreachable.
Reply from 172.16.2.1: Destination host unreachable.
Request timed out.

Ping statistics for 64.100.1.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
C:\>
C:\>
C:\>ping 64.100.1.2

Pinging 64.100.1.2 with 32 bytes of data:

Reply from 64.100.1.2: bytes=32 time=1ms TTL=126
Reply from 64.100.1.2: bytes=32 time=1ms TTL=126
Reply from 64.100.1.2: bytes=32 time=4ms TTL=126
Reply from 64.100.1.2: bytes=32 time=1ms TTL=126

Ping statistics for 64.100.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 4ms, Average = 1ms

C:\>
```

PC3 Command Prompt:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 64.100.1.2

Pinging 64.100.1.2 with 32 bytes of data:

Reply from 192.168.1.1: Destination host unreachable.
Reply from 192.168.1.1: Destination host unreachable.
Request timed out.
Reply from 192.168.1.1: Destination host unreachable.

Ping statistics for 64.100.1.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 64.100.1.2

Pinging 64.100.1.2 with 32 bytes of data:

Reply from 64.100.1.2: bytes=32 time=2ms TTL=125
Reply from 64.100.1.2: bytes=32 time=9ms TTL=125
Reply from 64.100.1.2: bytes=32 time=4ms TTL=125
Reply from 64.100.1.2: bytes=32 time=3ms TTL=125

Ping statistics for 64.100.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 9ms, Average = 4ms

C:\>
```